

arXiv:2504.03553

KnowSelf

Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

*Enabling agents to autonomously regulate knowledge utilization
through situational self-awareness*

Current Agent Training Uses 'Flood Irrigation'

Indiscriminate Knowledge Injection Hurts Efficiency

Three dominant paradigms — all blindly apply gold trajectories / feedback / knowledge at every step

Trajectory Imitation

ReAct, vanilla fine-tuning

Memorizes gold action sequences without situational judgment

Trial & Error Refinement

Reflexion, self-correction loops

Repeats failed attempts — costly, no external knowledge integration

Knowledge-Augmented

KnowAgent, WKM

Retrieves external KB at every step regardless of actual need

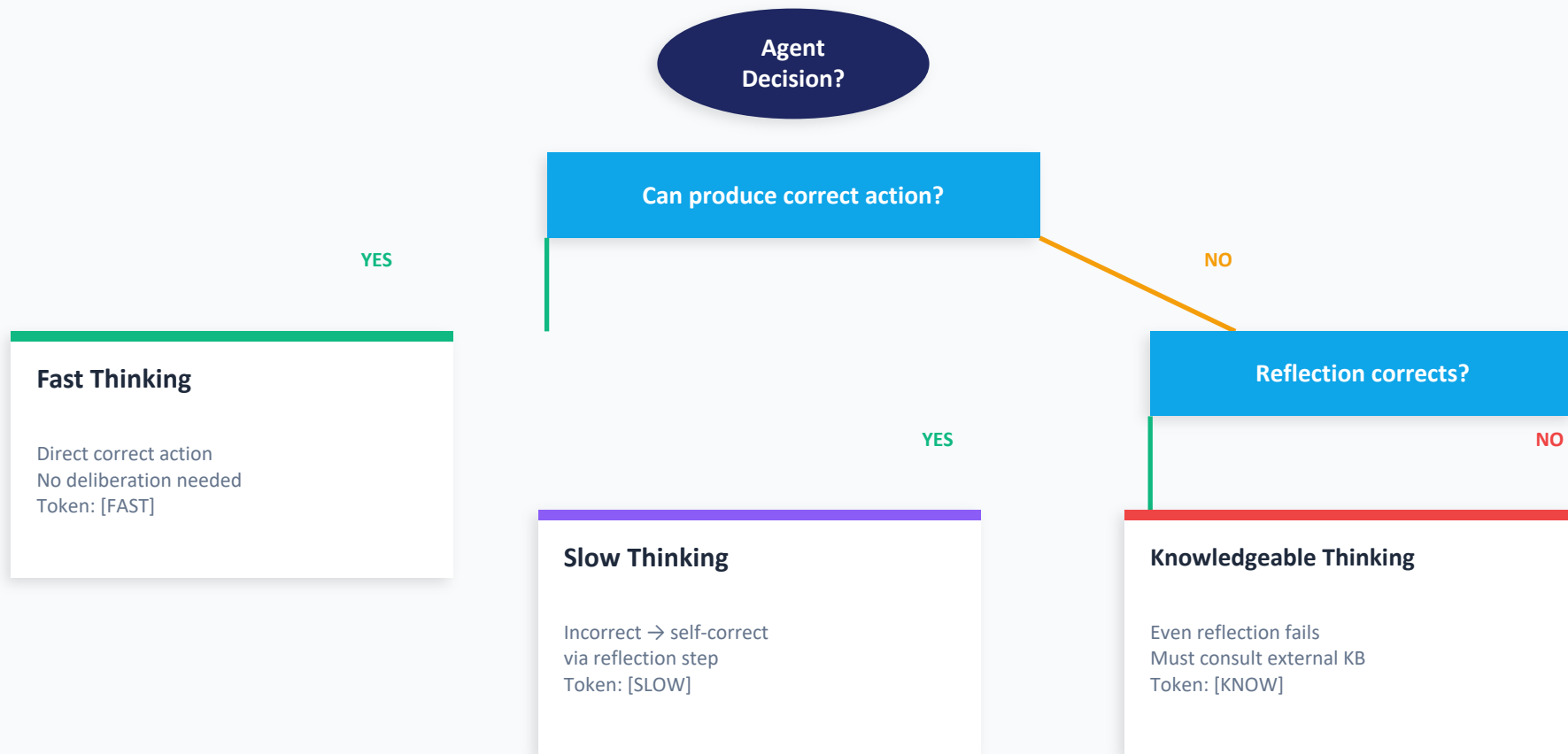


Result: Fragile planning on novel inputs · Excessive inference cost · Pattern collapse

Humans possess metacognitive self-awareness — KnowSelf brings this to LLM agents.

Three Situational Modes of Thinking

Fast · Slow · Knowledgeable



KnowSelf Framework

Data Construction → Two-Stage Training → Self-Aware Inference

1

Self-Awareness Data Construction

Agent self-explores environments

Heuristic marks situation type:

[FAST] — action matches gold

[SLOW] — reflection fixes it

[KNOW] — KB consultation needed

Special tokens inserted per step



2

Two-Stage Training

Stage 1 — SFT:

Supervised fine-tuning on
labeled self-explored data

Stage 2 — RPO:

Reward-based Policy Optimization
refines token generation policy



3

Self-Aware Inference

Agent generates special tokens:

[FAST] → direct action

[SLOW] → reflection loop

[KNOW] → KB retrieval

Contextual triggers, not rules

Selective knowledge utilization

Lightweight heuristic labeling — no expensive human annotation required

KnowSelf Correctly Handles Complex Tasks Where Baselines Fail

ALFWorld household task comparisons

Task A: Put a hot mug in cabinet

✗ OpenAI-O1

Opens microwave → incorrectly attempts to take an egg → fails to heat the mug before placing.

✓ KnowSelf (Llama-8B)

Generates [SLOW] token → reflection: "I need to heat the mug first." Uses open microwave → heats mug → places correctly.

Token: [SLOW] → Reflection resolves

Task B: Put a saltshaker in drawer

✗ DeepSeek-R1

Reflects "drawer is open" → tries to close it first → fails → gets stuck. Inappropriate reflection for a placement task.

✓ KnowSelf (Llama-8B)

Generates [KNOW] token → external KB: "place object without removing unrelated items." Correct spatial reasoning, no unnecessary reflection.

Token: [KNOW] → KB retrieval resolves

ALFWorld Results: KnowSelf Achieves 84.33% with Llama-8B

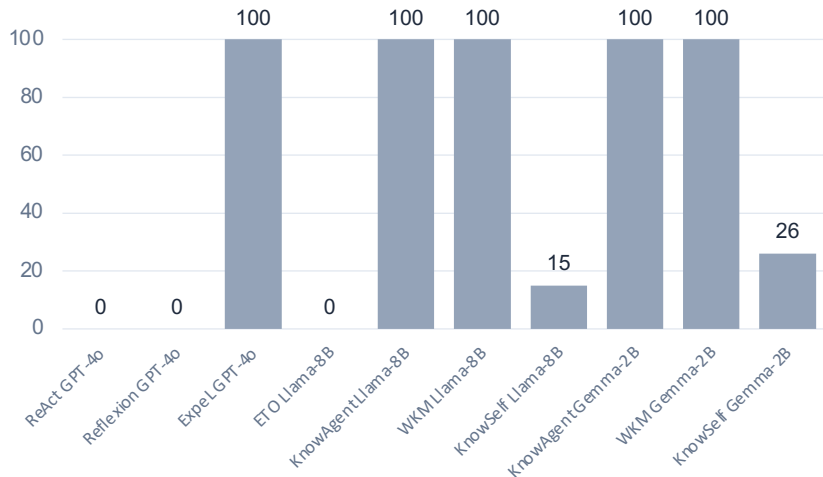
Success rates (%) across 6 task types · Know% = proportion of steps using external knowledge

Backbone	Method	Know%	Put	Clean	Heat	Cool	Exam	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
	KnowSelf	15%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
	KnowSelf	26%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

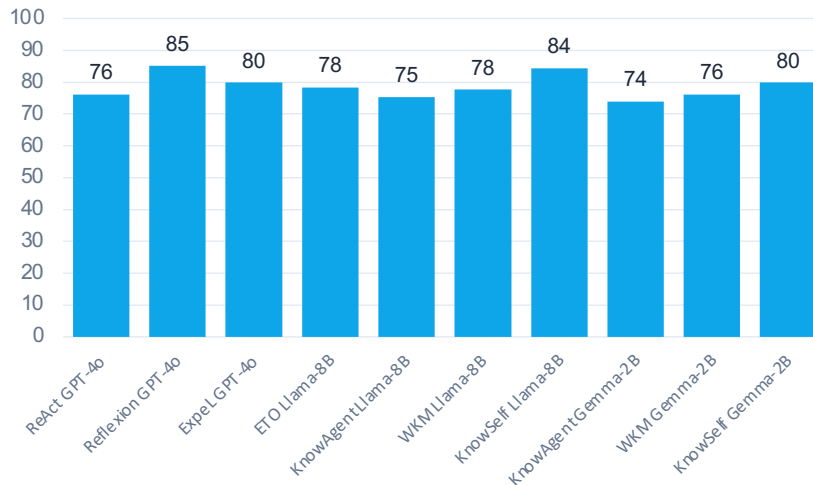
Minimal Knowledge, Maximum Impact

Knowledge usage (%) vs. overall accuracy on ALFWorld

External Knowledge Usage (%)



Overall Accuracy (%)



~15%

KnowSelf Llama-8B

knowledge usage

84.33% acc.

~26%

KnowSelf Gemma-2B

knowledge usage

79.85% acc.

100%

KnowAgent / WKM / Expel

knowledge usage

75–78% acc.

Key Takeaways

Self-Awareness Is the Missing Piece in Agent Planning

1

Indiscriminate knowledge injection wastes resources — methods using 100% knowledge (KnowAgent, WKM, ExpeL) consistently underperform KnowSelf's selective approach.

2

KnowSelf's three-situation taxonomy (fast / slow / knowledgeable thinking) provides a principled, human-inspired framework for agents to autonomously regulate cognitive resources.

3

Llama-8B + KnowSelf achieves 84.33% on ALFWorld — surpassing GPT-4o ReAct (76.12%) and GPT-4o ExpeL (79.85%) — using only 15% knowledge.

4

Special-token mechanism is lightweight: heuristic labeling on self-explored trajectories, no expensive human annotation.

5

Scales down effectively — Gemma-2B with KnowSelf reaches 79.85%, matching GPT-4o ExpeL and exceeding all other Gemma-2B baselines.